Prediction window

Mission control needs a position AND an honest range.

05 A

TASK 1 · CENTRAL PREDICTION

A probe starts at $x_0 = 12$ m and moves at $v = -2$ m/s for 4 s. $x_f = x_0 + v\,t =$ ____

TASK 2 · THE WINDOW

Start error ±1 m. Velocity error ±0.2 m/s. Largest error = start error + (velocity error × time) = ____ → window ____ m to ____ m

TASK 3 · MORE TIME

Redo the window for 10 s. What happens and why?

TASK 4 · UNITS FIRST

A train moves 40 m/s for 2 hours. Convert first, then find Δx in km.

ANALYST SIGN-OFF

A negative x does not mean a collision because ____. The window is part of the answer because ____.

R3 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every prediction

05 M

DIRECTIONS · Work each problem through all four boxes.

PROBLEM 1 · A train moves at 40 m/s for 2 hours. Find its displacement in km.

1 READ

v = ____ · t = ____ h = ____ s

2 DRAW

sketch the trip

3 SOLVE

$\Delta x = v \times t$ (units first!)

4 WRITE

The train moves ____ km because ____.

PROBLEM 2 · A cart starts at 10 ± 1 m and moves 2.0 ± 0.2 m/s for 5 s. Predict x and the range.

1 READ

x_0 = ____ · v = ____ · t = ____ · errors ____

2 DRAW

sketch the window

3 SOLVE

$x = x_0 + vt$ · error = $\delta x_0 + t \cdot \delta v$

4 WRITE

The cart will be at ____ m, between ____ and ____ m.

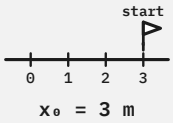
Hardest box: □ Read □ Draw □ Solve □ Write

R3 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: predictions

Language goal: I can **predict** with a range.



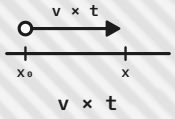
starting position

posición inicial · posição inicial

where it begins

my language: _____

use it: _____



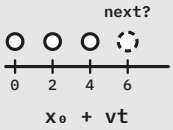
change

cambio · variação

how far it moves

my language: _____

use it: _____



predict

predecir · prever

say where it will be

my language: _____

use it: _____



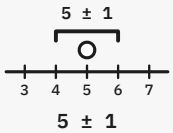
error

error · erro

how unsure we are

my language: _____

use it: _____



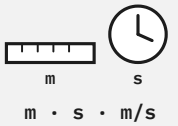
range

rango · intervalo

lowest to highest

my language: _____

use it: _____



units

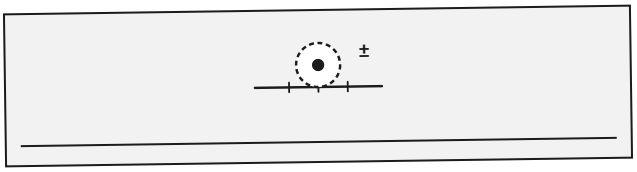
unidades · unidades

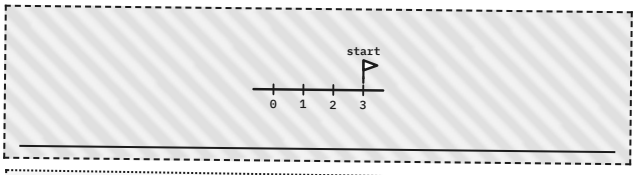
what the number measures

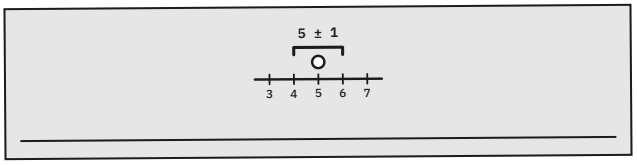
my language: _____

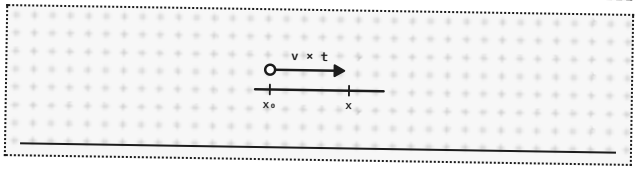
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner □

R3 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

VOL. 1 · NO. 5 · R3 REPORTER: _____

The 2026-XJ Daily

Prediction day: "Not a guess — a window"

BREAKING · HEADLINE (include a range!)

THE STORY

A test probe started at 30 ± 2 m and moved at −3.0 ± 0.1 m/s for 10 s. Report the central prediction AND the window.

BY THE NUMBERS

central x = ____
half-width = ____
window ____ to ____

LETTERS TO THE EDITOR

"The prediction crosses zero, so the probe must crash!" The editor replies:

SCIENCE DESK

Why does a longer prediction usually have a wider window?

CORRECTION BOX

A rival paper reported "Probe at exactly 0 m." Write the correction:

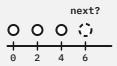
R3 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 5 · R3 · SEI · SENTENCE FRAMES

Say it, then write it

Language goal: I can **predict** with a range.



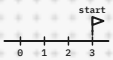
predict
predecir · prever
say what comes next



range
rango · intervalo
lowest to highest



error
error · erro
how unsure we are



starting position
posición inicial · posição inicial
where it begins

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: start · change · range · error · wider

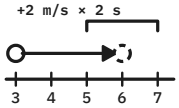
To predict, I take the ____ position and add the ____ (v × t). Every measurement has ____, so I give a ____. The longer the time, the ____ the range.

2 · SAY IT TO A PARTNER, THEN WRITE IT



It starts at ____ m and moves ____ m/s, so after ____ s it is at ____ m.
The range is ____ m to ____ m because ____.

3 · SOLVE WITH THE STEPS



Start at 3 m. Move +2 m/s for 2 s. Error ±1 m.

STEP 1 · CHANGE	STEP 2 · ADD START	STEP 3 · RANGE
____ × ____ = ____ m	____ + ____ = ____ m	____ m to ____ m

4 · EXPLAIN IT · CHOOSE A FRAME

☐ My prediction is ____ because ____.
☐ The range is important because ____.
☐ If the time doubles, the range ____ because ____.

I said it ☐ · I wrote it ☐

R3 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Unlock the code: predictions

05 C

DIRECTIONS · Solve each station. Copy the letter into the lock.

STATION 1

Start 2 m, +3 m/s for 4 s. Final x?

W 14 m M 12 m P 10 m

STATION 2

Start 8 m, −2 m/s for 3 s. Final x?

A 14 m I 2 m E −6 m

STATION 3

Prediction 6 ± 1 m. The range is…

N 5 to 7 m R 6 to 7 m

STATION 4

40 m/s for 2 h. First convert 2 h to…

D 7,200 s T 120 s

STATION 5

Start error ±1 m, v error ±0.5 m/s, 4 s. Largest error?

O ±3 m U ±1.5 m

STATION 6

More time usually makes the window…

W wider S narrower

LOCK [1] [2] [3] [4] [5] [6]

Hint: what we sketch around a prediction.

R3 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: a prediction is not a wild guess

05 CS

YOUR READER

A younger sibling who thinks a prediction is just a guess.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **BLUE** — Position marks with numbers and units
- ☐ ◌ **PURPLE** — A velocity arrow showing the car's steady velocity
- ☐ ◯ **PENCIL** — The assumption in a speech bubble: "if it keeps ____"

1 · THE DATA *A toy car at 0, 2, and 4 m at 0, 1, and 2 s.*

CAPTION _____

2 · THE PATTERN *A character spots how far it moves each second.*

CAPTION _____

3 · THE PREDICTION *Where will it be at 4 s? Show the prediction window.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **data · pattern · prediction · assumption**

R3 1 2 3

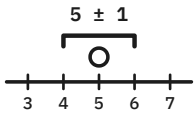
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · prediction range

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Prediction range · intervalo de predicción (ES) · intervalo de previsão (PT)



TODAY'S CASE

A cart starts at 2 m. Its steady velocity could be anywhere from +2 to +3 m/s for the next 2 s.

01 · SKETCH AND LABEL

Mark the start, then draw one blue position dot for each possible endpoint at the two speed limits.

02 · CHANGE THE PICTURE

Shade the interval between the dots. Label one finish inside the range and one outside.

Annotate your sketches. Circle the mark that best shows prediction range. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

My endpoints change if …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · prediction range

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Prediction range · intervalo de predicción (ES) · intervalo de previsão (PT)

A · The cart will certainly stop at the midpoint.

B · A forecast gives a lower and an upper endpoint.

01 · DEFINITION

What does prediction range mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else